Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

CH_3COOH dimerise in benzene according to reaction : $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

If 60% mole of CH_3COOH goes in dimer formation and remaining is in monomer form then what is average molecular mass of CH_3COOH.

(A) 90

(B) 60

(C) 85.71

(D) 96

Q.2

$KO_2 + H_2O \rightarrow KOH + H_2O_2 + O_2$.
$14.2gm.KO_2$ when treated with excess H_2O, it gives only $1.7gmH_2O_2$. So $\%$ yield of H_2O_2 should be -

(A) 20%

(B) 30%

(C) 40%

(D) 50%

Q.3

A chemist decided to determine the Empirical formula of an unknown compound. He collects following information :
(i) Compound contains $2:1$ ratio of H and O atoms (number of atoms)
(ii) Compound has 40% C by mass
(iii) Compound contains C, H and O only
What is the empirical formula of the compound -

(A) CH_3O

(B) CH_2O

(C) C_2H_2O

(D) CH_3O_2

Q.4

In the reaction, $2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$
(1) 6 L HCl (aq) is consumed for every 3 L $H_2(g)$ produced

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(2) $33.6\ \mathrm{L}\ \mathrm{H_2(g)}$ is produced regardless of temperature and pressure for every mole Al that reacts.
(3) $33.6\ \mathrm{L}\ \mathrm{H_2(g)}$ at STP is produced for every $27\mathrm{gm Al}$ that reacts.
(4) $11.2\ \mathrm{L}\ \mathrm{H_2(g)}$ at STP is produced for every $\mathrm{HCl(aq)}$ consumed.
Choose the correct options -

(A) 1,2

(B) 1,3

(C) 3,4

(D) $1, 2, 3, 4$

Q.5

Which of the following molarity values of ions in an aqueous solution of $5.85\%\mathrm{w/vNaCl}$, $5.55\%\mathrm{w/vCaCl_2}$ and $6\%\mathrm{w/vNaOH}$ are correct.
(1) $\left[\mathrm{Cl^{-1}}\right] = 2\mathrm{M}$
(2) $\left[\mathrm{Na^{+}}\right] = 1\mathrm{M}$
(3) $\left[\mathrm{Ca^{2+}}\right] = 0.5\mathrm{M}$
(4) $\left[\mathrm{OH^{-}}\right] = 1.5\mathrm{M}$
Choose the correct options -

(A) 1,2

(B) 2,3

(C) 3,4

(D) $1, 3, 4$

Q.6

A sample of $\mathrm{K_2CO_3}$ weighing 13.8 gram was treated by a series of reagents so as to convert 50% of its carbon to $\mathrm{K_2Zn_3[Fe(CN)_6]_2}$. How much Zn (in grams) (atomic mass of $\mathrm{Zn} = 65.5$) is required for the reaction?

(A) 0.20

(B) 3.1

(C) 0.82

(D) 0.54

Q.7

A gaseous mixture of propane and butane of volume 3 litre on complete combustion produces 11.0 litres of $\mathrm{CO_2}$ at standard temperature and pressure. The ratio of volume of butane to propane is:

(A) $1 : 2$

(B) $2 : 1$

(C) $2 : 3$

(D) $4 : 5$

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Q.8

Calculate the volume of H_2 at STP that will be displaced by 1 gram of Zn when it is completely dissolved in dilute sulphuric ac id.

(A) $0.1425 dm^3$

(B) $2.3425 dm^3$

(C) $0.3425 dm^3$

(D) $1.3425 dm^3$

Q.9

$5 mmol$ of SO_2Cl_2 is hydrolyzed completely to make a $100\ mL$ solution. Which of the following statements will apply appropriately? Assume no gases were allowed to escape out of solution.

(A) The solution would be 0.05 in H_2SO_4.

(B) The solution would be $0.05M$ in HCl.

(C) If $10\ mL$ of the stock solution is neutralized by $0.2M\ NaOH$, $10\ mL$ of this base solution would be required.

(D) If $10\ mL$, of the above solution is titrated with excess of $AgNO_3$, $1.5 mmol$ of $AgCl$ would be formed.

Q.10

You are given $5\ m$ (molal) aqueous solution of H_2SO_4 of density $1.49 gm/ml$, then -

(A) molarity (M) of the solution $= 5M$

(B) mole fraction of solute $= 0.083$

(C) \% by weight of $H_2SO_4 = 40\%$

(D) concentration of H^+ ion $= 10M$

Q.11

A sample of H_2O_2 solution labelled as 28 volume has density of $265\ g/L$. Mark the correct option(s) representing concentration of same solution in other units -

(A) $M_{H_2O_2} = 2.5$

(B) $\%\frac{W}{V} = 17$

(C) Mole fraction of $H_2O_2 = 0.2$

(D) $M_{H_2O_2} = 13.88$

Q.12

Passage:

We know that balancing of a chemical equation is entirely based on law of conservation of mass. However the

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concept of Principle of Atom Conservation (POAC) can also be related to law of conservation of mass in a chemical reaction. So, POAC can also act as a technique for balancing a chemical equation.

Question:

Which of the following relation is correct regarding the numerical coefficients $\mathrm{p}, \mathrm{q}, \mathrm{r}$ in the balanced chemical equation: $\mathrm{pA} + \mathrm{qB_2} \longrightarrow \mathrm{rA_2\ B_5}$

(A) $2\mathrm{p} = \mathrm{r}$

(B) $q = 1.25p$

(C) $r = 2q$

(D) $\mathrm{q} = 0.8\mathrm{p}$

Q.13

Passage:

We know that balancing of a chemical equation is entirely based on law of conservation of mass. However the concept of Principle of Atom Conservation (POAC) can also be related to law of conservation of mass in a chemical reaction. So, POAC can also act as a technique for balancing a chemical equation.

Question:

If the weight ratio of C and $\mathrm{O_2}$ present is $1:2$ and both of reactants completely consume and form CO and $\mathrm{CO_2}$. What would be the mole ratio of CO and $\mathrm{CO_2}$ in mixture.

(A) $11:7$

(B) $2:1$

(C) $1:1$

(D) $1:2$

Q.14

Passage:

We know that balancing of a chemical equation is entirely based on law of conservation of mass. However the concept of Principle of Atom Conservation (POAC) can also be related to law of conservation of mass in a chemical reaction. So, POAC can also act as a technique for balancing a chemical equation.

Question:

If the atomic masses of X and Y are 10 an
and 30 respectively, then the mass of $\mathrm{XY_3}$ formed when
$120\ \mathrm{g}$ of $\mathrm{Y_2}$ reacts completely with X is :
Reaction $\mathrm{X} + \mathrm{Y_2} \rightarrow \mathrm{XY_3}$

(A) $133.3\ \mathrm{g}$

(B) $200\ \mathrm{g}$

(C) $266.6\ \mathrm{g}$

(D) $400\ \mathrm{g}$

Q.15

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Passage:

The strength of H_2O_2 is expressed in several ways like molarity, normality, \% (w/V), volume strength, etc. The strength of " $10\ V$ " means 1 volume of H_2O_2 on decomposition gives 10 volumes of oxygen at STP or 1 litre of H_2O_2 gives 10 litre of O_2 at STP. The decomposition of H_2O_2 is shown as under : $H_2O_2(aq) \longrightarrow H_2O(\ell) + \frac{1}{2}O_2(g)$

H_2O_2 can acts a oxidising as well as reducing agent, as oxidizing agent H_2O_2 converted into H_2O and as reducing agent H_2O_2 converted into O_2 both cases it's n-factor is 2 .

$\therefore$ Normality of H_2O_2 solution $= 2\times$ Molarity of H_2O_2 solution

Question:

What is the molarity of " $11.2\ V$ " of H_2O_2 ?

(A) $1M$

(B) $2M$

(C) $5.6M$

(D) $11.2M$

Q.16

Passage:

The strength of H_2O_2 is expressed in several ways like molarity, normality, \% (w/V), volume strength, etc. The strength of " $10\ V$ " means 1 volume of H_2O_2 on decomposition gives 10 volumes of oxygen at STP or 1 litre of H_2O_2 gives 10 litre of O_2 at STP. The decomposition of H_2O_2 is shown as under : $H_2O_2(aq) \longrightarrow H_2O(\ell) + \frac{1}{2}O_2(g)$

H_2O_2 can acts a oxidising as well as reducing agent, as oxidizing agent H_2O_2 converted into H_2O and as reducing agent H_2O_2 converted into O_2 both cases it's n-factor is 2 .

$\therefore$ Normality of H_2O_2 solution $= 2\times$ Molarity of H_2O_2 solution

Question:

What is the percentage strength $(\%w/V)$ of $11.2\ VH_2O_2$?

(A) 1.7

(B) 3.4

(C) 34

(D) None of these

Q.17

Passage:

The strength of H_2O_2 is expressed in several ways like molarity, normality, \% (w/V), volume strength, etc. The strength of " $10\ V$ " means 1 volume of H_2O_2 on decomposition gives 10 volumes of oxygen at STP or 1 litre of H_2O_2 gives 10 litre of O_2 at STP. The decomposition of H_2O_2 is shown as under : $H_2O_2(aq) \longrightarrow H_2O(\ell) + \frac{1}{2}O_2(g)$

H_2O_2 can acts a oxidising as well as reducing agent, as oxidizing agent H_2O_2 converted into H_2O and as reducing agent H_2O_2 converted into O_2 both cases it's n-factor is 2.

$\therefore$ Normality of H_2O_2 solution $= 2\times$ Molarity of H_2O_2 solution

Question:

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20 mL of H_2O_2 solution is reacted with 80 mL of $0.05MKMnO_4$ in acidic medium then what is the volume strength of H_2O_2 ?

(A) 2.8

(B) 5.6

(C) 11.2

(D) None of these

Q.18

Match the column -

Column I	**Column II**
(A) 4.1g H_2SO_3	(p) 200ml of 0.5 base is used for complete neutralization
(B) 4.9g H_3PO_4	(q) 200 milli moles of oxygen atoms
(C) 4.5g oxalic acid ($H_2C_2O_4$)	(r) Central atom has its highest oxidation number
(D) 5.3g Na_2CO_3	(s) May react with an oxidising agent
	(t) 100 milli moles of hydrogen atoms

(A) A-p,s,t; B-q,r ; C- p,q,s,t ; D-r

(B) A-t,q; B-r ; C- t,p,q,s ; D-p,q,s

(C) A-p,q,s; B-p,r,q ; C- p,q,r,s ; D-p,q,s

(D) A-r,s; B-p,r ; C- p ; D-p,q,s

Q.19

In the following reaction, ratio of x and y is $xHI + yHNO_3 - NO + I_2 + H_2O$.

Q.20

In the reaction $CO + O_2 \rightarrow CO_2; N_2 + O_2 \rightarrow NO$
10ml of mixture containing CO and N_2 required 7ml oxygen to form CO_2 and NO, on combustion. The volume of N_2 in the mixture will be _____ ml.

Answers & Solutions

Q.1 Answer:

85.71

Solution:

	$2CH_3COOH$	$\rightleftharpoons$	$(CH_3COOH)_2$
At $1 = 0$	1		x
At equilibrium	$1 - 2x$		0.3

Given $2x$ is 60% of 1

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$2x = \frac{60}{100} \times 1$

$x = 0.3$

$1 - 2(0.3) = 1 - 0.6 = 0.4$

$\bar{M} = \frac{0.4 \times 60 + 0.3 \times 120}{0.7} = \frac{24+36}{0.7} = \frac{60}{0.7} = \frac{600}{7} = 85.71$

Q.2 Answer:

50%

Solution:

$2KO_2 + 2H_2O \rightarrow 2KOH + H_2O_2 + O_2$

$$\text{mass of } H_2O_2 = 3.4\text{gm}$$

$\frac{14.2}{71}\text{gm} = 0.2 \text{ mole}$

$\frac{\text{moles of } KO_2}{2} = \frac{\text{moles of } H_2O_2}{1}$

$\frac{0.2}{2} = \frac{\text{moles of } H_2O_2}{1}$

$\%\text{ yield } = \frac{\text{actual}}{\text{theoretical}} \times 100 = \frac{1.7}{3.4} \times 100 = 50\%$

$\text{moles of } H_2O_2 = 1$

Q.3 Answer:

CH_2O

Solution:

$C \quad H \quad O$

$40\% \qquad x\% \qquad 60 - x\%$

$\frac{40}{12} \qquad \frac{x}{1} \qquad \frac{60-x}{16}$

$\text{Given : } \quad x : \frac{60-x}{16} = 2 : 1$

$\frac{x}{\frac{60-x}{16}} = \frac{2}{1}$

$16x = 120 - 2x$

$x = \frac{120}{18} = \frac{20}{3}$

$C \quad H \quad O$

$1 : 2 : 1$

Q.4 Answer:

3,4

Solution:

Given that, $2Al(s) + 6HCl$ **(aq)** $-2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(\ g)$

For each mole of Al **reacted,** 1.5 **mole of** H_2 **is formed, and for each mole of** $HCl(aq)$ **consumed,** 0.5

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mole or $11.2\ \mathrm{L}$ of $\mathrm{H_2}$ at STP formed.

Q.5 Answer:

$1, 3, 4$

Solution:

Only single solution have all these means $100\mathrm{ml}$ solution have $5.85\mathrm{gm NaCl} = 0.1$ mole and $5.55\mathrm{gm CaCl_2} = 0.05$ mole

$$\text{and } 6\mathrm{gm NaOH} = 0.15 \text{ mole}$$
$$\left[\mathrm{Cl^-}\right] = \frac{(0.1+0.05\times 2)\times 100}{100} = 2\mathrm{M}$$
$$\left[\mathrm{Na^+}\right] = \frac{(0.1+0.15)\times 1000}{100} = 2.5\mathrm{M}$$
$$\left[\mathrm{Ca^{2+}}\right] = \frac{0.05}{100} \times 1000 = 0.5\mathrm{M}$$
$$\left[\mathrm{OH^-}\right] = 1.5\mathrm{M}$$

Q.6 Answer:

0.82

Solution:

$\mathrm{No_2}$ of $\mathrm{K_2CO_3}$ mole $= \frac{13.8}{138} = \frac{1}{10}$

no of mole of $\mathrm{C} = \frac{1}{10}$,

Carbon which converted into $\mathrm{K_2Zn_3\left[Fe(CN)_6\right]_2} = \frac{1}{20}$ **mole**

$\therefore$ **12 mole of** C **required 3 mole of** Zn.

$\therefore$ **1/20 mole of** C **required** $\frac{3}{12} \times \frac{1}{20}$ **mole of** Zn.

mass of $\mathrm{Zn} = \frac{3}{12} \times \frac{1}{20} \times 65.5 = 0.819\mathrm{gram} \simeq 0.82\ \mathrm{g}$

Q.7 Answer:

$2 : 1$

Solution:

Suppose, the volume of propane = V L

$$\mathrm{C_3H_8(\ g) + 5O_2(\ g) \rightarrow 3CO_2(\ g) + 4H_2O(l)}$$
$$\mathrm{C_4H_{10}}(g) + \frac{13}{2}\mathrm{O_2}(g) \rightarrow 4\mathrm{CO_2}(g) + 5\mathrm{H_2O}(l)$$
$$(3-V) \quad \tfrac{13}{2}(3-V) \quad 4(3-V)$$

Total volume of $\mathrm{CO_2}$ **produced** $= 11\ \mathrm{L}$;

$3V + 4(3-V) = 11$

$\therefore V = 1$

$\therefore$ **volume of butane** $= (3-1) = 2L$

Thus, the ratio of volume of butane to propane $= 2 : 1$

Q.8 Answer:

$0.3425\mathrm{dm^3}$

Solution:

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$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

1 atom 1 molecule 1 molecule 1 molecule

1 gram-atom

1 mole

65.4 g $\quad 2$gm **or** 22.4dm^3

$\because$ 65.4 g **of** Zn **displaces** 2 g **of Hydrogen**

$\therefore$ 1.0 g **of** Zn **displaces** $\frac{2}{65.4} \times 1 = 0.0306$ g **of** H_2

$\because$ 65.4 g **of** Zn **displaces** 22.4dm^3 **of** H_2 **at S.T.P.**

$\therefore$ 1.0 g **of** Zn **displaces** $\frac{22.4}{65.4} \times 1.0 = 0.3425\text{dm}^3$

Q.9 Answer:

The solution would be 0.05 in H_2SO_4.

Solution:

The hydrolysis reaction is

$$\underset{5\text{mmol}}{SO_2Cl_2} + 2H_2O \longrightarrow \underset{5\text{mmol}}{H_2SO_4} + \underset{10\text{mmol}}{2HCl}$$

Molarity $(H_2SO_4) = \frac{5}{100} = 0.05\text{M}(A)$

Molarity $(HCl) = \frac{10}{100} = 0.10\text{M}(B$ **is incorrect** $)$

10 mL **stock solution has**

$= 10 \times 0.05 \times 2\,(H_2SO_4) + 10 \times 0.10(HCl)$

= 2 meq of acid

Volume of $NaOH$ **(0.2 M) required for** 10 mL **stock solution** $= \frac{2}{0.2} = 10$ mL **(c).** 10 mL **of stock solution has** $1\text{mmol}HCl$ **and would produce** $1.0\text{mmol}AgCl$ **on treatment with excess** $AgNO_3$**. (D is incorrect).**

Q.10 Answer:

molarity (M) of the solution $= 5\text{M}$

Solution:

(A) Total moles of solute $(H_2SO_4) = 5$

Total mass of solute $(H_2SO_4) = 5 \times 98 = 490\text{gm.}$

Volume of the solution $= \frac{\text{Total mass of solution}}{\text{Density of solution}} = \frac{490+1000}{1.49} = 1000\text{ml}$

Molarity of the solution $= \frac{\text{mole of solute}}{\text{volume of solution}} \times 1000 = \frac{5}{1000} \times 1000 = 5\text{M}H_2SO_4$

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(B) Mole fraction of the solute $= \frac{5}{5+\frac{1000}{18}} = \frac{5}{5+55.55} = 0.083$

Total mass of solution $x + 1000 = 5 \times 98 + 1000$

Q.11 Answer:

$M_{H_2O_2} = 2.5$

Solution:

$V_{\text{strength}} = 56,$

$$\therefore \quad M = \frac{28}{11.2} = 2.5$$
$$\therefore \quad 1\text{ L contain } 2.5 \text{ moles of } H_2O_2$$
$$\text{or} \quad 2.5 \times 34 = 85\text{ g} H_2O_2$$

wt. of 1 litre solution $= 265\text{ g} (\because d = 265\text{ g/L})$

$$\therefore \quad W_{H_2O} = 180\text{ g or moles of } H_2O = 10$$
$$x_{H_2O_2} = \frac{2.5}{2.5+10} = 0.2$$
$$\%\frac{W}{v} = \frac{2.5\times34}{1000} \times 100 = 8.5$$
$$m = \frac{2.5}{180} \times 1000 = 13.88$$

Q.12 Answer:

$q = 1.25p$

Solution:

On applying POAC for A and B respectively, we get

$1 \times n_A = 2 \times n_{A_2B_5} \quad \ldots(1)$

$2 \times n_{B_2} = 5 \times n_{A_2B_5} \quad \ldots(2)$

By eq. (1) and (2), $\frac{n_A}{4} = \frac{n_{B_2}}{5} = \frac{n_{A_2B_5}}{5}$

$\therefore$ Balanced equation : $4\text{ A} + 5\text{ B}_2 \rightarrow 2\text{ A}_2\text{ B}_5$

$\therefore \quad q = 1.25p$

On applying POAC for X and Y respectively, we get

$1 \times n_X = 1 \times n_{XY_3} \quad \ldots(3)$

$2 \times n_{Y_2} = 3 \times n_{XY_3} \quad \ldots(4)$

By eq. (3) and (4), $n_{Y_2} = 1.5 \times n_X$

By eq. (3) and (4), n

$\frac{n_X}{2} = \frac{n_{Y_2}}{3} = \frac{n_{XY_3}}{2}$

$\therefore$ Balance equation: $2X + 3Y_2 \rightarrow 2XY_3$

$n_{Y_2} = \frac{120}{60} = 2 \quad \therefore n_{XY_3} = \frac{4}{3}$

$\therefore \quad m_{XY_3} = \frac{4}{3} \times 100 = 133.3\text{ g}$

$\underset{w}{C} + \underset{2w}{O_2} \rightarrow \underset{x}{CO} + \underset{y}{CO_2}$

POAC for C $\quad \frac{w}{12} = x + y$

POAC for O $\quad \frac{2w}{32} \times 2 = x + 2y$

$y = \frac{w}{8} - \frac{w}{12} = \frac{w}{24}; x = \frac{w}{12} - \frac{w}{24} = \frac{w}{24}$

$x : y = 1 : 1$

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Q.13 Answer:

1 : 1

Solution:

On applying POAC for A and B respectively, we get

$$1 \times n_A = 2 \times n_{A_2B_5}$$
$$2 \times n_{B_2} = 5 \times n_{A_2B_5}$$
$$\text{By eq. (1) and (2), } \frac{n_A}{4} = \frac{n_{B_2}}{5} = \frac{n_{A_2B_5}}{5}$$
$$\therefore \quad \text{Balanced equation : } 4\,A + 5\,B_2 \rightarrow 2\,A_2B_5$$
$$\therefore \quad q = 1.25p$$

On applying POAC for X and Y respectively, we get

$$1 \times n_X = 1 \times n_{XY_3}$$
$$2 \times n_{Y_2} = 3 \times n_{XY_3}$$
$$\text{By eq. (3) and (4), } n_{Y_2} = 1.5 \times n_X$$

$$\text{By eq. (3) and (4), } n$$
$$\frac{n_X}{2} = \frac{n_{Y_2}}{3} = \frac{n_{XY_3}}{2}$$

$$\therefore \quad \text{Balance equation: } 2X + 3Y_2 \rightarrow 2XY_3$$
$$n_{Y_2} = \frac{120}{60} = 2 \quad \therefore n_{XY_3} = \frac{4}{3}$$
$$\therefore \quad m_{XY_3} = \frac{4}{3} \times 100 = 133.3\ g$$
$$C \quad + \quad O_2 \quad \rightarrow \quad CO + CO_2$$
$$w \quad 2wx\ mole \quad y\ mole$$
$$\text{POAC for C} \quad \frac{w}{12} = x + y$$
$$\text{POAC for O} \quad \frac{2w}{32} \times 2 = x + 2y$$
$$y = \frac{w}{8} - \frac{w}{12} = \frac{w}{24}; x = \frac{w}{12} - \frac{w}{24} = \frac{w}{24}$$
$$x : y = 1 : 1$$

Q.14 Answer:

133.3 g

Solution:

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On applying POAC for A and B respectively, we get

$$1 \times n_A = 2 \times n_{A_2B_5}$$
$$2 \times n_{B_2} = 5 \times n_{A_2B_5}$$
$$\text{By eq. (1) and (2), } \frac{n_A}{4} = \frac{n_{B_2}}{5} = \frac{n_{A_2B_5}}{5}$$
$$\therefore \quad \text{Balanced equation : } 4\,A + 5\,B_2 \rightarrow 2\,A_2B_5$$
$$\therefore \quad q = 1.25p$$

On applying POAC for X and Y respectively, we get

$$1 \times n_X = 1 \times n_{XY_3}$$
$$2 \times n_{Y_2} = 3 \times n_{XY_3}$$
$$\text{By eq. (3) and (4), } n_{Y_2} = 1.5 \times n_X$$

$$\text{By eq. (3) and (4), } n$$
$$\frac{n_X}{2} = \frac{n_{Y_2}}{3} = \frac{n_{XY_3}}{2}$$

$$\therefore \quad \text{Balance equation: } 2X + 3Y_2 \rightarrow 2XY_3$$
$$n_{Y_2} = \frac{120}{60} = 2 \quad \therefore n_{XY_3} = \frac{4}{3}$$
$$\therefore \quad m_{XY_3} = \frac{4}{3} \times 100 = 133.3\ g$$
$$C \quad + \quad O_2 \quad \rightarrow \quad CO + CO_2$$
$$w \quad 2wx\ mole \quad y\ mole$$
$$\text{POAC for C} \quad \frac{w}{12} = x + y$$
$$\text{POAC for O} \quad \frac{2w}{32} \times 2 = x + 2y$$
$$y = \frac{w}{8} - \frac{w}{12} = \frac{w}{24}; x = \frac{w}{12} - \frac{w}{24} = \frac{w}{24}$$
$$x : y = 1 : 1$$

Q.15 Answer:

1M

Solution:

(i) 1 litre of H_2O_2 (aq) provide 11.2 litre of O_2 at STP moles of $O_2 = \frac{11.2}{22.4} = 0.5\ n_{H_2O_2}$ required $= 0.5 \times 2$

$$M_{H_2O_2} = \frac{n_{H_2O_2}}{V_{\text{solution}}} = 1M$$

(ii) Strength in percentage means how many $g H_2O_2$ present per 100 mL

$\because$ $M \Rightarrow 1$ **and mol. wt. of** $H_2O_2 = 34$

$\therefore$ $3.4\ g H_2O_2$ **present per litre solution or** $3.4\ g H_2O_2$ **present per** 100 mL **of solution.**

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(iii) m-eq. of H_2O_2 = m-eq. of $KMnO_4$

$$20 \times N = 0.05 \times 5 \times 80 \Rightarrow N = 1$$

$$N = \frac{\text{volume strength of } H_2O_2}{5.6}$$

$$\Rightarrow \quad \text{Volume strength of } H_2O_2 = 5.6$$

Q.16 Answer:

3.4

Solution:

(i) 1 litre of H_2O_2 (aq) provide 11.2 litre of O_2 at STP moles of $O_2 = \frac{11.2}{22.4} = 0.5\ n_{H_2O_2}$ required = 0.5×2

$$M_{H_2O_2} = \frac{n_{H_2O_2}}{V_{\text{solution}}} = 1M$$

(ii) Strength in percentage means how many $g H_2O_2$ present per 100 mL

$\because$ $M \Rightarrow 1$ **and mol. wt. of** $H_2O_2 = 34$

$\therefore$ $3.4\ g H_2O_2$ **present per litre solution or** $3.4\ g H_2O_2$ **present per** 100 mL **of solution.**

(iii) m-eq. of H_2O_2 = m-eq. of $KMnO_4$

$$20 \times N = 0.05 \times 5 \times 80 \Rightarrow N = 1$$

$$N = \frac{\text{volume strength of } H_2O_2}{5.6}$$

$$\Rightarrow \quad \text{Volume strength of } H_2O_2 = 5.6$$

Q.17 Answer:

5.6

Solution:

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Q.18 Answer:

A-p,s,t; B-q,r ; C- p,q,s,t ; D-r

Solution:

(A) $4.1\ g H_2SO_3 = \frac{1}{20}$ **mole** $= \frac{1}{10}$ **equivalent and** $200 \times 0.5 = 100$ **m equi.** $= 0.1$ **equ. base**

$\overset{+4}{H_2SO_3} +$ oxidising agent $\longrightarrow \overset{+6}{H_2SO_4}$

(B) $4.9\ g H_3PO_4 = \frac{1}{20}$ **mole** $= \frac{3}{20}$ **equi.** $= \frac{1}{20} \times 4000$ **milli mole**

O **atom** $= 2000$ **milli moles atom.**

Oxidation state $= +5$

(C) 4.5 g **oxalic acid** $(H_2C_2O_4) = \frac{1}{20}$ **mole** $= \frac{1}{10}$ **equivalent** $= 200$ m **mole** O **atom**

$= 100$ m **mole** H **atom**

$H_2C_2O_4$ + **oxidising agent** $\longrightarrow 2CO_2 + 2H^+$.

(D) $5.3gm. = \frac{1}{20}$ **mole** $= \frac{1}{10}$ **equivalent with acid.**

Oxidation number $= +4$.

Q.19 Answer:

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Solution:

The balance equation is $6HI + 2HNO_3 \rightarrow 2NO + 3I_2 + 4H_2O$

Q.20 Answer:

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Solution:

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$ $\qquad$ $N_2(g) + O_2(g) \rightarrow 2NO(g)$

x $\quad$ $\frac{x}{2}$ $\qquad\qquad$ y $\quad$ y

$x + y = 10$ **...(1)**

$\frac{x}{2} + y = 7$ **...(2)**

From $1\&2 \quad x = 6, y = 4$

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